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Three arguments for abstraction in phonology

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Much recent research in cognitive and brain sciences links speech processing to the auditory and articulatory systems. These facts notwithstanding, here, I argue that phonology is abstract, algebraic, and substance-free. I first review challenges to this hypothesis and explain why it remains a viable possibility. I next suggest that an algebraic, substance-free phonology confers adaptive benefits, and finally, I support this proposal by evidence from three case studies. Using the restrictions on onset structure (e.g., *bnif* > *bdif* > *lbif*), I show that the phonological constraints on syllable structure are inexplicable by auditory and articulatory pressures; hence, they are *abstract*. A second case, from identity restrictions, suggests that phonological constraints are *algebraic*, as speakers generalize them to features that are unattested in their language; this is the case in both Hebrew and American Sign Language (ASL). Finally, I show that sign-language-naïve speakers—both adults and infants—freely extend their knowledge of spoken-language phonology to ASL signs, suggesting that phonological knowledge is partly *amodal*. I conclude that phonological grammar is substance-free.



When asked why certain sound patterns are preferred to others—why, for instance, people *blog*, not *lbog*—many scholars appeal to the phonetic substance. *Lbog*, they argue, is disliked because its phonetic substance is harder to articulate and decode from the speech signal (e.g., Ohala 1983; Hayes et al. 2004; Pulvermüller et al. 2005). Phonology, in this view, is *substance-based*. Phonological representations encode the analog and continuous properties of acoustic and articulatory substance (e.g., voice onset time, amplitude, gesture overlap). Furthermore, those aspects of the “substance” aren’t just “sitting there”, passively. Rather, the phonetic substance plays a *causal* role in the generation of phonological patterns. Phonological patterns, then, arise not from abstract rules and constraints but solely from bodily action—that of the ear and mouth.

In this paper, I argue against this position. To do so, I first consider recent developments in brain and cognitive science that seemingly support the substance-based view. The facts, to be clear, are indisputable. But, in my view, they do not challenge the possibility that phonology is substance-free; the second section motivates this assertion. Finally, I present three arguments that phonology is indeed abstract, algebraic, and amodal, and support them by extensive experimental evidence. My conclusions converge with the other papers in this volume and elsewhere in the literature (Andersson 2025; Chabot 2025; Gorman & Reiss 2025; Hale & Reiss 2008; Reid et al. 2017; Shen 2025; Tanaka et al. 2011; Volenec & Reiss 2020) to suggest that phonology is *substance-free* (Berent 2013; Hale & Reiss 2008).

1 Substance-free phonology: still a viable hypothesis

Linguists have long noticed that phonological patterns aren’t fully arbitrary. Rather, marked phonological structures typically pose greater demands on the articulatory and motor system (e.g., Stampe 1973; Hayes et al. 2004). These observations could suggest that synchronic phonological processes are guided by sensorimotor pressures, rather than by abstract rules. Phonology, in that view, is substance-based

To illustrate the view, consider, again, restrictions on onset structure (e.g., *bla* > *lba*). In an influential paper, John Ohala and Haruko Kawasaki-Fukimori (1990) argued that syllable structure ought to be captured not by abstract phonological restrictions on sonority, but by acoustic phonetic parameters, such as amplitude, periodicity, spectral shape and fundamental frequency. Recent evidence from neuroscience seems to vindicate this proposal.

To evaluate the role of phonetic substance, Organian and Chang (2019) implanted electrodes in patients’ brains and traced their responses to spoken language. Results showed that areas in the superior temporal gyrus record various acoustic correlates of phonological structure. Moreover, computational analyses confirmed that those acoustic properties are sufficient to recover syllable structure and lexical stress. Their results would seem to imply that, for the human brain, syllable structure is substance-based.

Other results link phonology to the articulatory system. In this account, the preference for unmarked syllables (e.g., *ba* > *bla*) arises from the desire to optimize the parallel transmission of consonants and vowels (Kawasaki-Fukumori 1992; Mattingly 1981; Wright 2004). In line with this proposal, the evidence from neuroscience suggests that articulatory action occurs covertly and, as such, may constrain speech perception even when people do not engage in overt talking. The idea is that, to perceive speech, people must act: they must *simulate* what it would be for them to articulate the relevant speech sound, and in so doing, they engage both the relevant articulators (e.g., lips) as well as the motor areas that control them (Fadiga et al. 2002; Liberman et al. 1967).

One piece of evidence comes from the role of the speech motor system in speech perception. It is well known that primary motor cortex is topographically organized—certain areas of M1 control action by specific bodily organs, from the leg and hand all the way to the articulatory motor system, including the lips and tongue (e.g., Pulvermüller et al. 2006; for review Badino et al. 2014; Skipper et al. 2017). Thus, as people produce labial sounds, they engage the lip motor area (more than the tongue motor area), whereas the production of coronal sounds engages the tongue motor area (more than the lip; e.g., Pulvermüller et al. 2006). Crucially, those articulatory motor regions also charge selectively as people *listen* to speech sounds. Hearing *ba*, for instance, elicits stronger activation of the lip motor area, whereas hearing *da* selectively charges the tongue motor area (e.g., Pulvermüller et al. 2006).

This activation, moreover, is not epiphenomenal but causal: the activation of the motor system is necessary for speech perception to take place. This much becomes clear from experiments that selectively disrupt activation in these motor regions, using a technique called transcranial magnetic stimulation (TMS). TMS temporarily disrupts activity in specific brain regions (e.g., the lip motor area) by altering the brain's electromagnetic activity (e.g., Pascual-Leone et al. 1993). In so doing, one can gauge whether the area of interest, such as the lip motor area, is necessary for the encoding of labial speech sounds. If it is, then its disruption via TMS ought to selectively impair the perception of labial sounds relative to controls (e.g., coronal sounds). This is precisely what the results show (e.g., D'Ausilio et al. 2012; D'Ausilio et al. 2009; Möttonen & Watkins 2009; Smalle et al. 2014).

Other experiments find the same when motor activity is disrupted mechanically—by having participants bite on their lips, tongues, or on an object that engages these articulators. Results suggest that these manipulations selectively impair the perception of the relevant speech sounds. For example, when people lightly bite on their lip, they are less able to identify labial sounds (e.g., Berent et al. 2020b). Likewise, when young infants bite on a teething toy that engages their tongue, they lose their ability to perceive coronal speech sounds (Bruderer et al. 2015).

Together, these results suggest that as we listen to spoken language, we tacitly engage in motor action—we simulate what it would “be like” to articulate those speech sounds. Given

that (as shown above), linguistic structure can be directly extracted from the acoustic input, and given further that the brain response to this acoustic information engages motor action, it would appear that phonology is subsumed by acoustic and articulatory “substance”; there is no abstract, substance-free phonology in human brains.

These challenges notwithstanding, I believe the hypothesis of substance-free phonology remains utterly plausible. To be clear, this is not to say that the previously cited results aren’t real. Human brains do encode the substance of speech—that much is undeniable. But the positive evidence for the coding of phonetic substance does not show that the brain *doesn’t* encode abstract algebraic structure. Thus, the evidence discussed is perfectly compatible with the hypothesis that human brains encode speech along two parallel processing streams. One stream registers continuous, analog aspects of speech substance; another encodes abstract, substance-free phonological structure.

To be clear, the evidence I will present is based primarily on behavioral experiments. As such, this evidence suggests a *functional* distinction between the substance-based and substance-free streams; their precise neural implementations remain unknown. But given that the functional cognitive architecture arises from the organization of the brain, it is quite likely that these streams correspond to two distinct networks in human brains.

2 Substance-free phonology is adaptive

Why bother, you wonder? If the brain already registers phonetic detail, why worry about abstraction? Furthermore, if the brain does encode abstraction, why does phonological structure often mimic phonetic form? For example, why does the sonority cline of the syllable recapitulate the peaks and valleys of acoustic modulation and articulatory aperture? In short, why be wasteful? If the brain encodes phonetic substance, why bother forming abstraction alongside the phonetic detail?

In what follows, I suggest that substance-free, algebraic phonology confers adaptive benefits. Nonetheless, an adaptive phonological system ought to also be “mindful” of the sensorimotor limitations on the human body: its outputs should be readily articulated and perceived by the human mouth and ear. Phonological structure, I suggest, has evolved to satisfy these two competing constraints. Let me explain these claims.

2.1 Algebraic phonology is adaptive

To appreciate why a substance-free phonology is adaptive, it is helpful to consider the role of phonology within the language system. All sides agree that language is productive and that an adaptive phonological system must therefore support linguistic productivity. Natural systems, however, form new elements by *two* distinct methods—blending and combination (Abler 1989). So, in principle, either method could guide phonological productivity.

Blending systems generate new forms by mixing existing substances, akin to baking a cake from flour, sugar, and butter, or creating the color green by mixing blue and yellow. In both cases—the cake and the color green—a new instance is formed by blending existing substances. Blending systems, then, can certainly yield new forms, and the ingredients do matter—omitting the sugar from one’s cake certainly makes a noticeable difference. Still, once blended, the individual ingredients interact and are no longer recognizable in the whole (e.g., you cannot distinguish the bits of sugar in your cake, or the blue in the green).

But in **combinatorial** systems, you can. These systems generate new forms by combining discrete elements. DNA, for instance, combines four bases—adenine (A), thymine (T), guanine (G), and cytosine (C). Like the blending system, combinatorial systems are productive. For example, the four DNA bases can combine productively to form new life forms. To attain productivity, however, the building blocks must remain discrete and unaltered when combined. Indeed, it is precisely the sequencing of discrete DNA bases that guides gene expression, regulation and the formation of new proteins. In combinatorial systems, then, productivity is attained by a system of “discrete infinity” (Chomsky 2017).

Language is arguably a combinatorial system, and this is also the case for phonology. To support discrete infinity in language, phonology, then, must minimally combine discrete elements without blending. And, to attain the full power of linguistic productivity, the system must further exhibit systematicity, and compositionality (Fodor & Pylyshyn 1988), akin to how algebra forms powerful expressions by operating on the constituent structure of discrete, abstract variables (Marcus 2001). By hypothesis, then, phonology is not only substance-free but *algebraic* (Berent 2013).

But phonetic substance is neither discrete nor combinatorial; phonetic cues are, by definition, analog and continuous, and they are known to interact (e.g., Miller & Volaitis 1989; Keating 1988). Phonetic substance, then, fails to “deliver the goods” necessary to attain the discrete infinity of language. And it is for this reason that I claim that phonology must be substance-free.

2.2 Phonetically-sensible phonology is adaptive

Being substance-free, however, does not entail that phonological patterns are phonetically arbitrary. To be transmitted and perceived by the human body, phonological structure must be ultimately expressed in phonetic form. And the processing of phonetic form is surely constrained by the human body. Certain phonetic properties simply won’t register by the human sensorimotor system, and thus, they won’t survive the “telephone game” that underlies the cultural evolution of language (Blevins 2004). Phonology, then, cannot afford to be utterly arbitrary—it must favor phonological structures that can be expressed in “sensible” phonetic forms.

This, to be clear, does not mean that every phonological process is “phonetically sensible”. Indeed, several authors have identified systematic phonological alternations that violate

sensorimotor pressures. Examples of such “crazy rules” include the preference of Buriat stems with front vowels to take suffixes with back vowels (as opposed to the expected harmony on frontness/backness; Chabot 2021), and the unexpected role of *m* as a trigger for realization of dental stops as fricatives in Attic Greek (Sommerstein 1973). Thus, the pressure for “phonetically sensible” output is clearly a soft violable constraint, not an inviolable “must”. Still, all things being equal, one would not expect such “phonetically senseless” processes to be prevalent.

2.3 Algebraic optimization

Phonology, then, must abide by two conflicting pressures. To survive cultural evolution, its outputs must be phonetically sensible, as “senseless” phonological outputs are less likely to persist across generations. But to attain productivity, the phonological grammar must also be algebraic and substance-free. How can the system satisfy these two conflicting constraints?

The answer, I suggest, is *not at once*. The phonological system can abide by each of these pressures as long as each such constraint is met at a different time scales. Phonetic constraints apply *diachronically*, during language evolution (via natural selection and cultural evolution), and it is these substance-based restrictions that give rise to “phonetically sensible” constraints, such as a preference for sonority rises. But at the *synchronic* level, within the brains and minds of individual speakers, phonological representations and grammatical processes are algebraic and substance-free.

Phonology, then, works much like the blind watchmaker in Dawkins’ account of natural selection (Dawkins 1987). In this analogy, the watchmaker is the phonological grammar, and its creations are phonological representations (the clocks). As the watchmaker assembles its phonological “clocks”, each one of its actions is algebraic and substance-free, and so are its outputs. But unbeknownst to the clockmaker, the plan it follows is not arbitrary; its distinct algebraic steps were selected to render the phonological “clock” resilient to phonetic pressures. When seen from afar, it becomes evident that the actions and their sequences make “phonetic sense”. And yet, the immediate causes of each of the watchmaker’s actions—of each computation within the phonological grammar—are strictly algebraic and substance-free.

Thus, at the synchronic level, phonological representations and causes are strictly algebraic—the sensorimotor substance plays no direct causal role in phonological computations; this aspect of the proposal agrees with the substance-free hypothesis advanced by Hale and Reiss (2008). Unlike Hale and Reiss, however, in my proposal, sensorimotor “substance” could nonetheless play a role in shaping the design of phonological systems diachronically and phylogenetically. I refer to this design as *algebraic optimization* (Berent 2013). The seemingly bipolar nature of phonology, then, is not a bug, but an adaptive feature. This is precisely what would be expected given its function in the language system.

3 Phonology is substance-free: three arguments

What evidence is there to support this proposal? Since the phonological literature is full of evidence for the phonetic grounding of phonology, in this short piece, I will not seek to defend that aspect of the proposal. Instead, I strive to show that phonological constraints—the immediate causes of phonological computation—are irreducible to sensorimotor demands: they are abstract, algebraic, and amodal, hence, substance-independent.

3.1 Phonology is irreducible to sensorimotor demands

To show that phonology is irreducible to sensorimotor demands, consider, again, the restrictions on syllable structure. Language surveys show that onsets with large rises in sonority are preferred to small rises, which, in turn, are preferred to plateaus; least preferred are sonority falls (*blif* > *bnif* > *bdif* > *lbif*; Berent et al. 2007, data from Greenberg 1978; on codas showing the mirror preference for falls, see Bat-El 2012; Clements 1990; Maïonchi-Pino & Runge 2024.). Furthermore, a large body of literature shows that people are sensitive to the syllable hierarchy, even when all syllable types are unattested in their language: the better-formed the syllable, the more faithfully it is encoded. These preferences have been reported among adult speakers of various languages (e.g., Berent et al. 2007; Berent 2008; Zhao & Berent 2016; Tamasi & Berent 2015), children (e.g., Berent et al. 2011; Jarosz 2017; Ohala 1999; Berent et al. 2008; Berent et al. 2012a), and even newborn infants (Gómez et al. 2014). **Figure 1A** presents representative data from adult English speakers (data from Berent et al. 2007; Experiment 1).

These facts can be readily captured by the hypothesis that *all* speakers possess universal restrictions on syllable structure (e.g., Prince & Smolensky 1993/2004). The nature of these restrictions, however, is controversial. One view asserts that the constraints on syllable structure are *abstract*. Whether or not these abstract constraints concern a (substance-free) sonority feature specifically or other features (c.f., Clements 1990; Smolensky 2006) is immaterial; either way, the constraint is substance-free.

Critically, this proposal further predicts that syllables that violate this constraint are *repaired* by the **grammar** (e.g., *lbif* → *lebif*). Since the formal analysis is provided elsewhere (Berent et al. 2009), here, I will not delve into the details. My goal is to underscore the logical argument: misidentification does not necessarily arise from a “substance problem”.

I say “not necessarily” because *some* forms of misidentification can undoubtedly result from “mis-hearing” (i.e., failure to encode the auditory or phonetic form) or “mis-talking” (an articulation failure) and “mis-talking” could occur even when people passively listen to speech (since speech perception engages the articulatory system, as shown in the previous section). Substance-failures, then, can certainly contribute to repair. Critically, however, they aren’t its *sole* cause. In my proposal, repair (and thus, misidentification) could also arise from grammatical

constraints that are algebraic. Paradoxically, then, misidentification is the hallmark of a substance-free computation.

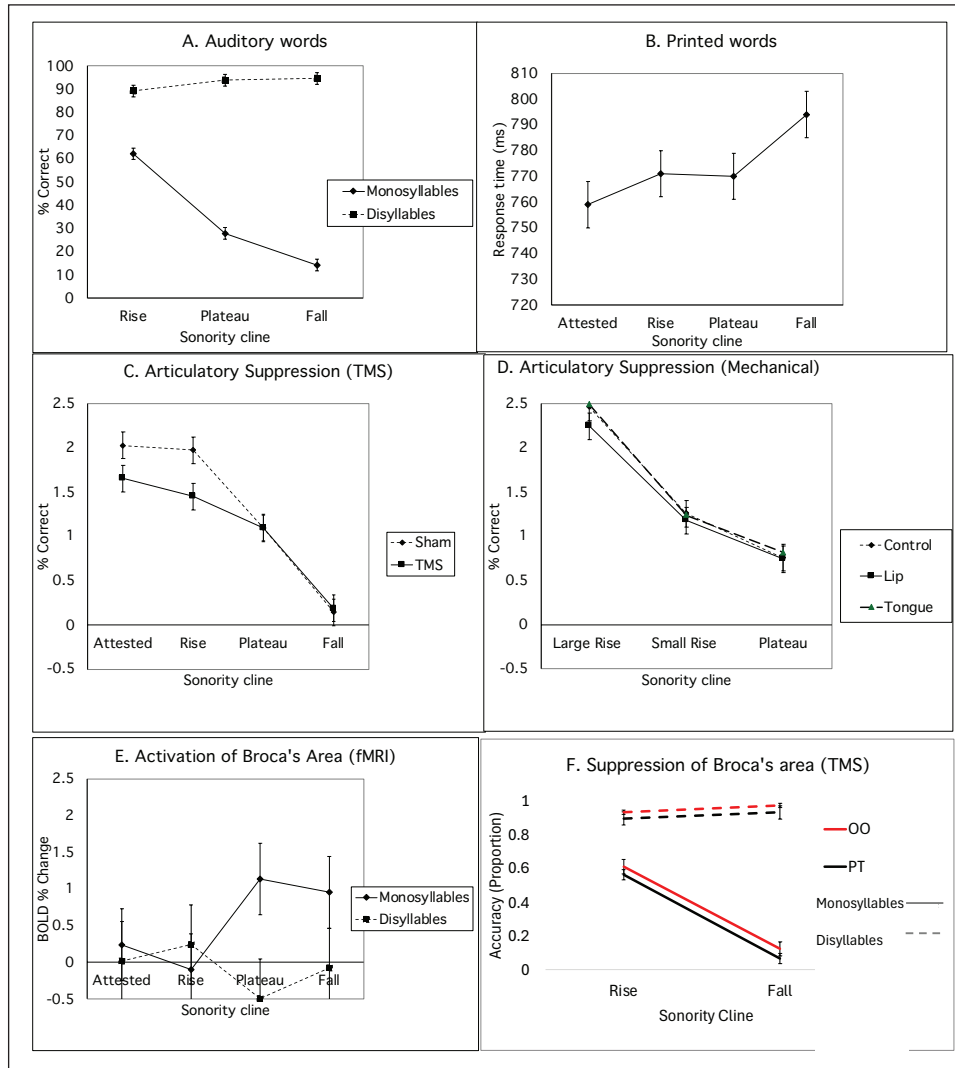


Figure 1: The restriction on syllable structure. Marked monosyllables are harder to identify when the materials are either spoken (A) or printed (B; shown are responses to non-identical trials, e.g., *lbif-lebif*). These difficulties persist when the contribution of covert articulation is disrupted, by applying TMS to the lip motor area (relative to a sham condition; C) or via mechanical suppression (i.e., pressing on the lips, tongues, or Control; D). While auditory and articulatory constraints cannot explain the restriction on the syllable hierarchy (A-D), abstract grammatical constraints do, as the syllable hierarchy engages Broca's area (BA 45; E) and the disruption of this region (specifically, the pars triangularis, PT) disrupts syllable count (relative to the disruption of the lip motor area, the orbicularis oris, OO). Data from Berent et al. 2007 (A); Berent & Lennertz 2010(B); Berent et al. 2015(C); Berent & Platt 2022 (D); Berent et al. 2014a(E); Berent et al. 2023(F).

To support this proposal, my lab has systematically examined various substance-based explanations for repair and showed that they cannot fully account for misidentification. Let me briefly consider these explanations and the counter-evidence, in turn.

a. Repair does not only arise from “mishearing”. To evaluate the auditory/phonetic account, one can examine whether people remain sensitive to syllable structure when the experimental materials are presented in a printed form—in the absence of any auditory or phonetic demands. At first blush, reading seems irrelevant to phonology. But a large literature shows that people do extract phonological structure in silent reading (e.g., Perfetti et al. 1992; Van Orden et al. 1990). Building on this literature, we thus expect participants to decode the phonological structure of the syllables. The critical question is whether people will remain sensitive to their syllable structure. Since the phonological representations of printed words do not require hearing, evidence for the repair of printed words cannot possibly arise from a failure to encode the acoustic input, as the auditory/phonetic account asserts¹. Are marked syllables, then, still repaired in print?

To find out, participants were presented with two successive nonwords, either identical (e.g., *lbif*-LBIF), or nonidentical and epenthetically related (e.g., *lbif*-LEBIF); the task is to determine whether the two items are identical. If marked syllables are typically misidentified due to a simple auditory/phonetic failure, then now that the materials are printed (and thus, impose no auditory/phonetic demands), misidentification ought to vanish. But the results show that it doesn't: marked syllables still take longer for readers to decode from print. **Figure 1B** provides evidence from printed words, modeled after the spoken items in 1A (Berent & Lennertz 2010; similar results are reported in Berent et al. 2009; Lennertz & Berent 2015; Tamasi & Berent 2015). These results suggest that the demands exacted by these syllables aren't auditory.

b. Repair is not solely due to “mis-talking”. Could these difficulties arise from articulatory reasons? Could *lbif* be taxing because, as people perceive speech, they attempt to covertly articulate it? To evaluate this idea, one can disrupt the articulatory system. If the troubles with *lbif* (and other marked syllables) arise from a failed articulatory attempt, then once articulation is disrupted, the demands of marked syllables should vanish; persistence would suggest that a source that is substance-free.

One set of experiments disrupted the articulatory motor system in the brain (specifically, the lip motor area, the orbicularis oris) using transcranial motor stimulation; **Figure 1C** presents the results (from Berent et al. 2015). Other experiments disrupted articulation mechanically, by

¹ Could readers still extract phonetic forms from print? They certainly can (Abramson & Goldinger 1997). But this fact does not undermine my argument. The “mishearing” account, recall, states that repair arises from failures to encode the auditory *input* (speech). The repair of printed words counters this argument, as the input to reading is visual (not auditory). So, if people still repair printed words, then repair cannot arise from “mis-hearing”. Critically, this would remain the case even if readers converted visual input to phonetic detail; phonetics, here, is *output*, not input, so this possibility is irrelevant to my argument.

having people bite on either their lips or tongues as they heard either labial or coronal stimuli. **Figure 1D** presents syllable count responses to labial stimuli (e.g., *plik*, *pnik*, *ptik*) as participants bite on their lips, tongue, or do nothing (Control; data from Berent & Platt 2022; for similar results, see Zhao & Berent 2013). In all cases, sensitivity to the syllable hierarchy persisted.

This, to be sure, is not because the articulatory manipulations were ineffective. Recall that the perception of *phonetic* contrasts is exquisitely sensitive to articulatory manipulations (as discussed previously). Indeed, when articulatory suppression was applied in a *phonetic* task (the perception of voicing continua), speech perception was affected in the expected, selective manner: the disruption of lip articulation selectively disrupted the perception of labial continua more than coronal continua, and this was the case when suppression was applied either mechanically, by biting on the lips (Berent et al. 2020b) or by disrupting the lip motor area via TMS (Berent et al. 2023). This confirms that *phonetic* processing is highly dependent on articulatory simulation. Phonology, however, is not.

A neuroimaging study further bears this conclusion out (Berent et al. 2014a). Here, participants engaged in a syllable-counting task while their brains were scanned using fMRI. If syllable structure arises from sensorimotor demands, then as syllable structure becomes more marked, activation in the auditory and motor areas ought to increase. This was not the case. Instead, activation increased in Broca's area (specifically, the Pars Triangularis, PT)—a primary language hub. Thus, as a monosyllable became worse-formed, activation in the PT increased (for results, **Figure 1E**).

A final TMS experiment confirmed that this activation of Broca's area isn't just epiphenomenal. Rather, Broca's area plays a causal role in the computation of syllable structure. To gauge the role of Broca's area in the syllable hierarchy, one can disrupt its activation by TMS and examine its effect on a syllable-counting task. If the activation of Broca's area is necessary for the computation of syllable structure, then once it's disrupted, the extraction of syllable structure should be impaired; this is exactly what was found: participants became biased to identify all stimuli as disyllabic, regardless of whether they were monosyllables or disyllables (Berent et al. 2023; see **Figure 1F**). Summarizing, the results show that while phonetics is clearly sensitive to acoustic and articulatory substance, phonology is not.

3.2 Phonology is algebraic

If phonology *isn't* based on sensorimotor substance, then what *is* it based on? What principles drive computations in the phonological grammar? Above, I suggested these principles are algebraic and substance-free. We now turn to test this hypothesis.

a. What are algebraic rules? One critical hallmark of algebraic operations is that they generalize across the board. Consider, for example, a constraint on identical elements, such as *XX. In this

constraint, *X* is a variable that stands for an entire *class* of phonological elements, such as “any feature” or “any syllable”. Expressions such as **XX* are akin to mathematical algebraic expressions ($y = 2X$, where *X* is “any integer”), insofar as both apply to entire classes (e.g., any integer; any syllable), as opposed to their specific instances (e.g., 3, “ba”). Since, by definition, these classes do not specify phonetic substance, it follows that if the phonological mind is algebraic, then it is also *substance-free*. But is it?

Upon a cursory inspection, it appears that it is, as identity restrictions (in phonology), and reduplication (in morphophonology) are frequent across languages—spoken (Suzuki 1998; Walter 2007) and signed (Wilbur 2009); in what follows, I will use the term “*doubling*” to refer to identity and reduplication, collectively. And, doubling, as noted, is widely attested across languages. Still, the fact that one can *describe* doubling in algebraic terms does not prove that human brains work that way. And indeed, on an alternative account, what would appear to present an algebraic restriction could arise from the co-occurrence of specific instances. For example, a preference for *dada* does not concern reduplication per se (i.e., *XX* where *X* stands for any syllable). Rather, people only track the statistical co-occurrence of specific syllable *tokens* (e.g., *baba*, *dada*), as they demonstrably can (Saffran et al. 1996). It is the statistical frequency of specific phonological substances that drive their preference, not algebraic structure.

To adjudicate between these explanations, one can examine whether people still constrain doubling (*XX*) when *XX* and *XY* forms include elements with novel feature values—features that are unattested in one’s language. These test items challenge statistical models for two reasons. First, in these cases, the frequency of both *XX* and *XY* forms is vanishingly low, so they are statistically indistinguishable from each other. Second, generalization to unattested features is known to pose serious challenges for statistical computational models (Marcus 1998; Berent et al. 2012b). If phonology only tracks the frequency of specific phonological tokens, then the distinction between *XX* and *XY* forms should not extend to unattested features. In contrast, if the ban is substance-free and algebraic, then people could still contrast them by virtue of their structure.

b. Evidence for algebraic phonological rules. To test this prediction, participants were presented with identical elements comprising feature values that are utterly unattested in their language. This test was applied in two language modalities—spoken (Hebrew) and signed (ASL).

Consider first the case of Hebrew. Like all Semitic languages, Hebrew bans **XXY* roots, but allows *YXX* forms. Thus, *dideb* is banned, whereas *bided* (he isolated) is attested. To determine whether this restriction concerns reduplication per se (as opposed to the statistical infrequency), one can examine whether Hebrew speakers extend this restriction to *th* (/θ/)—a phoneme that is unattested in Hebrew, and its place feature—tongue tip constriction area ‘wide’ (Gafos 1999) is likewise unattested. Hebrew speakers, however, systematically favored *kathath* to *thathak* forms (Berent et al. 2002).

Another set of experiments tested the same in ASL. To this end, these experiments compared the responses of native ASL signers to novel XX and XY forms, whose handshape feature was unattested in ASL. Thus, both forms (XX and XY) included two unattested features, but in the XX items, these features were identical. Since ASL has a productive reduplicative morphology (Wilbur 2009), signers are expected to prefer novel reduplicative signs over non-reduplicative ones. The critical question was whether they would extend this preference to unattested features.

Results showed that they do, and this was the case in both offline rating (Berent et al. 2014b) as well as in online procedures, such as a lexical decision study (Berent et al. 2014b) and even the Stroop task (Dupuis & Berent 2015). Moreover, control experiments showed that nonsigners exhibited no such preference. In fact, nonsigners showed a reliable doubling *aversion* (i.e., $XX < XY$). Below, we discuss this aversion in detail, but for now, we can conclude that the reduplication preference of *signers* arises from their knowledge of language, and not from sensorimotor factors alone. Summarizing, then, these results show that speakers of reduplicative morphologies—Hebrew and ASL—each project their phonological knowledge to novel features. These results suggest that phonology is an algebraic, and thus, substance-free.

3.3 Phonology is amodal

So far, we have shown that phonological restrictions are substance-free and, for this reason, they can be freely extended to novel features. All these demonstrations, however, are bound by language modality. Hebrew speakers, for instance, generalize to novel words in their spoken language, whereas ASL signers project generalizations to novel manual signs. In principle, however, the scope of algebraic phonology could be even broader. An algebraic restriction, such as *XX (where X is “any syllable”), can theoretically apply to *any* syllable—spoken or signed. So, if phonology is algebraic, people might be able to generalize doubling restrictions from their spoken language to a sign language, and do so spontaneously, despite having no command of a sign language.

Before we evaluate this prediction, we must first address two challenges. To constrain the identity of signed syllables, sign-naïve speakers must first be able to identify syllables in a sign language. And of course, they must also possess the requisite doubling constraints and apply them within their L1 (e.g., English). Can speakers rise to these challenges?

In what follows, I first show that indeed, they do. I next proceed to examine amodal projection by adults and infants.

a. Speakers extract syllables from ASL signs. Sign-naïve speakers can track the number of syllables in ASL signs (contrasting monosyllables and disyllables, akin to the English contrast between *pen* and *pencil*) and distinguish them from the number of morphemes (akin to the contrast between *pen* and *pens*; Berent et al. 2013). Speakers can readily do so because, like

spoken syllables, signed syllables are marked by sonority peaks, such as movement (Sandler & Lillo-Martin 2006), which are phonetically salient to sign-naïve speakers. Interestingly, sign-naïve speakers spontaneously expect signed syllables to be marked by sonority peaks, akin to the structure of syllables in their spoken language, and it is this amodal principle that explains their ability to extract syllables from signs (Berent et al. 2013). Given that signed syllables are salient to speakers, we can now return to the critical question of whether they further constrain their identity.

b. English speakers demonstrably constraint phonological identity and reduplication in English. As noted, in languages like Hebrew and ASL, doubling is parsed as reduplication ($\{X\}X_c$), and it is systematically *preferred*. In English, by contrast, XX has no systematic morphological function. Thus, XX in forms like *paper* presents phonological identity, and identity is banned (by the Obligatory Contour Principle; Leben 1973; McCarthy 1979). It follows that the doubling (generally, XX) is structurally ambiguous, as it is amenable to two competing parses—phonological or morphological. When XX is parsed by the phonology, as identity, XX is banned. But when the same surface form is parsed morphologically, as reduplicative ($\{X\}X_c$), XX is now systematically preferred (Inkelas & Zoll 2005; Berent et al. 2016).

Before we tested generalizations to ASL signs, we first wished to ascertain that English speakers apply these doubling restrictions to novel *English* words. Results showed that when presented with bare phonological forms, English speakers indeed systematically disprefer doubling to controls (e.g., *slaflaf* < *slafmat*: Berent et al. 2016; Berent et al. 2017; and *panana* < *panaka*: Berent et al. 2021a). But when the exact same surface forms were presented as reduplication (e.g., *slaf* = one ball; *slaflaf* = a set of balls), doubling was now preferred (e.g., *slaflaf* > *slafmat*: Berent et al. 2016; Berent et al. 2017; and *panana* > *panaka*: Berent et al. 2021a).

c. Amodal projections. The results discussed thus far suggest that English speakers (a) possess the requisite grammatical knowledge of doubling; and (b) they can extract the requisite constituents (syllables) from ASL signs. With these results in hand, we can now turn to the critical question: can English speakers spontaneously project grammatical constraints from the grammar of their (spoken) language to signs?

To find out, a series of experiments presented speakers who are sign-native with novel ASL signs—either XX and XY, in two contrastive contexts: phonological or morphological (**Figure 2A**). In the phonological context, people saw a photo of a novel object and were asked to choose which ASL sign makes a better name for it—XX or XY; repetition, then, had no bearing on meaning, so it is purely phonological, i.e., identity.

In the morphological condition, by contrast, repetition conveyed a systematic change in meaning, such as plurality. To this end, people were presented with a base X, along with an object (e.g., a ball); next they saw either a set of identical objects (e.g., a set of three balls), to

suggest semantic plurality; the task was to choose which of two signs—XX or XY—is a better name for that set in ASL (**Figure 2A**).

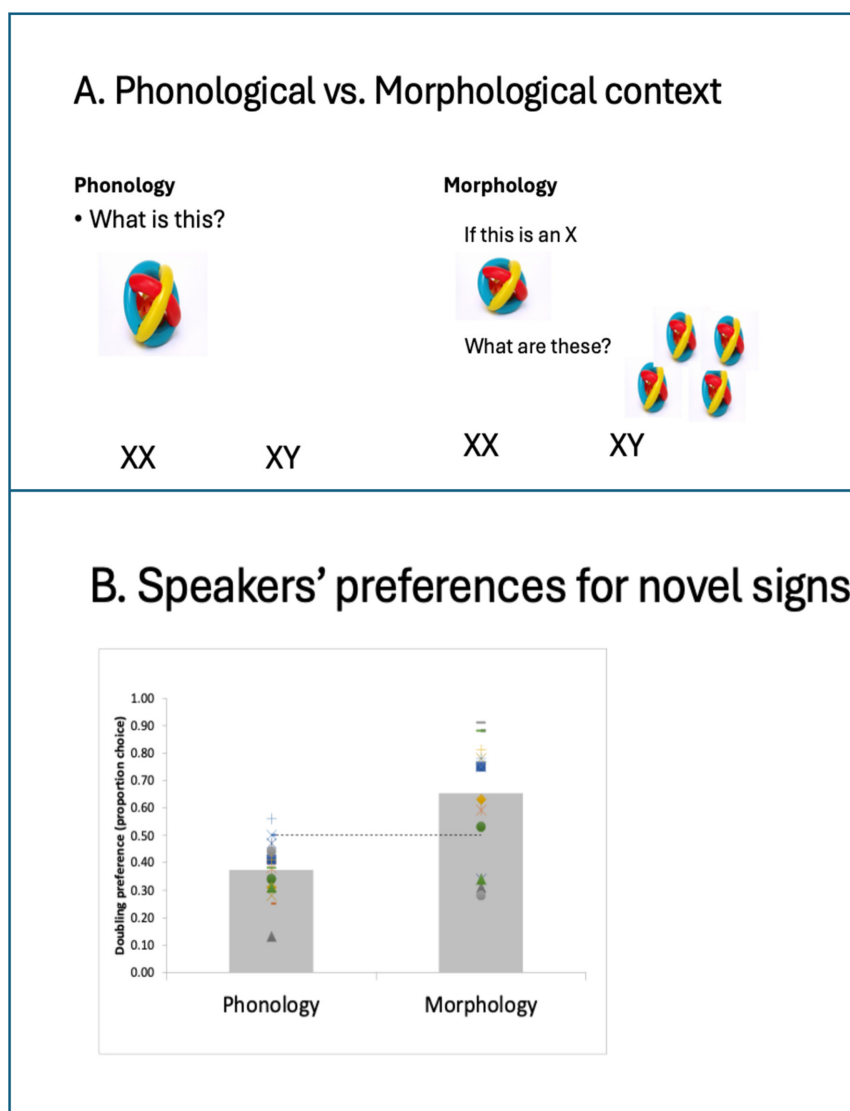


Figure 2: Cross-modal generalizations. Panel A illustrates the presentation of XX and XY signs in a phonological vs. morphological context; Panel B presents the responses of English speakers to doubling in novel signs; each symbol captures the mean doubling preference of an individual participant; the dotted line captures chance level (.5); data from Berent et al. 2016.

Results showed that the doubling preference shifted across conditions. In the phonological condition, people dispreferred identity (i.e., $XX < XY$), in line with the ban on identity in their spoken language phonology (Berent et al. 2016; Berent et al. 2021a). But when the context

suggested morphological reduplication, they preferred reduplicative signs (i.e., $XX > XY$; **Figure 1B**)).

It is unlikely that these preferences only arise from nonlinguistic strategies, such as iconicity. First, iconicity does not explain why doubling is banned, in the phonological context. Second, iconicity cannot explain the doubling preference in the morphological context. This is the case because speakers' preference for ASL signs demonstrably depends on the morphological structure of their *spoken* language. English speakers, as noted, preferred XX signs to express semantic plurality, which is productively marked by their L1 morphology, whereas speakers of Mandarin did not—in line with the absence of plural morphology in their spoken language (Berent et al. 2020a). Further evidence for the role of linguistic constraints is provided by the types of semantic relations expressible by reduplication. In Hebrew morphology, reduplication productively expresses diminution (e.g., *kelev-klavalav*; 'dog-puppy'), whereas in English, it does not. Correspondingly, Hebrew speakers preferred XX signs when the context suggested diminution (X = a ball; XX = a small ball; Berent et al. 2016) whereas English speakers did not (Berent et al. 2016). Together, these results demonstrate that phonological structure is not only substance-free but modality-free.

Similar conclusions emerge when doubling is explored in the brains of young infants. A large body of literature suggests that young infants (Marcus et al. 1999; see also Bouchon et al. 2015; Gerken 2006, 2010; Gerken et al. 2015; Gervain et al. 2012; Gervain et al. 2008; Gervain & Werker 2013; Marcus et al. 2007; Wagner et al. 2011) can track the abstract algebraic structure of identity, distinguish it from nonidentical elements (e.g., XX vs. XY) and constrain its location (e.g., XXY vs. XYX); indeed, this ability is present at birth (Gervain et al. 2012). Unknown, however, is whether young infants who have had no previous experience with a sign language can project these restrictions to ASL signs (for preliminary conflicting results, see Rabagliati et al. 2012; Rabagliati et al. 2019).

To find out, researchers presented 6-month-old infants who were sign-naïve with novel ASL signs, either XX and XY , and gauged their brain response using near-infrared spectroscopy (NIRS)—a technique that tracks hemodynamic response by measuring the absorption of infrared light by the brain (Berent et al. 2021b).

To determine whether responses to signs arise from their linguistic structure or from visual characteristics alone, the experiment further compared the responses to signs with matched visual controls—cartoons of trees, fitted with hand-shaped leaves, such that the spatial and temporal properties precisely matched the configuration of the signers' body (torso and hands) and its movement. In the experiment, participants were presented with blocks of XX and XY stimuli (either ASL signs or visual controls), and their brain response was monitored (**Figure 3a**).

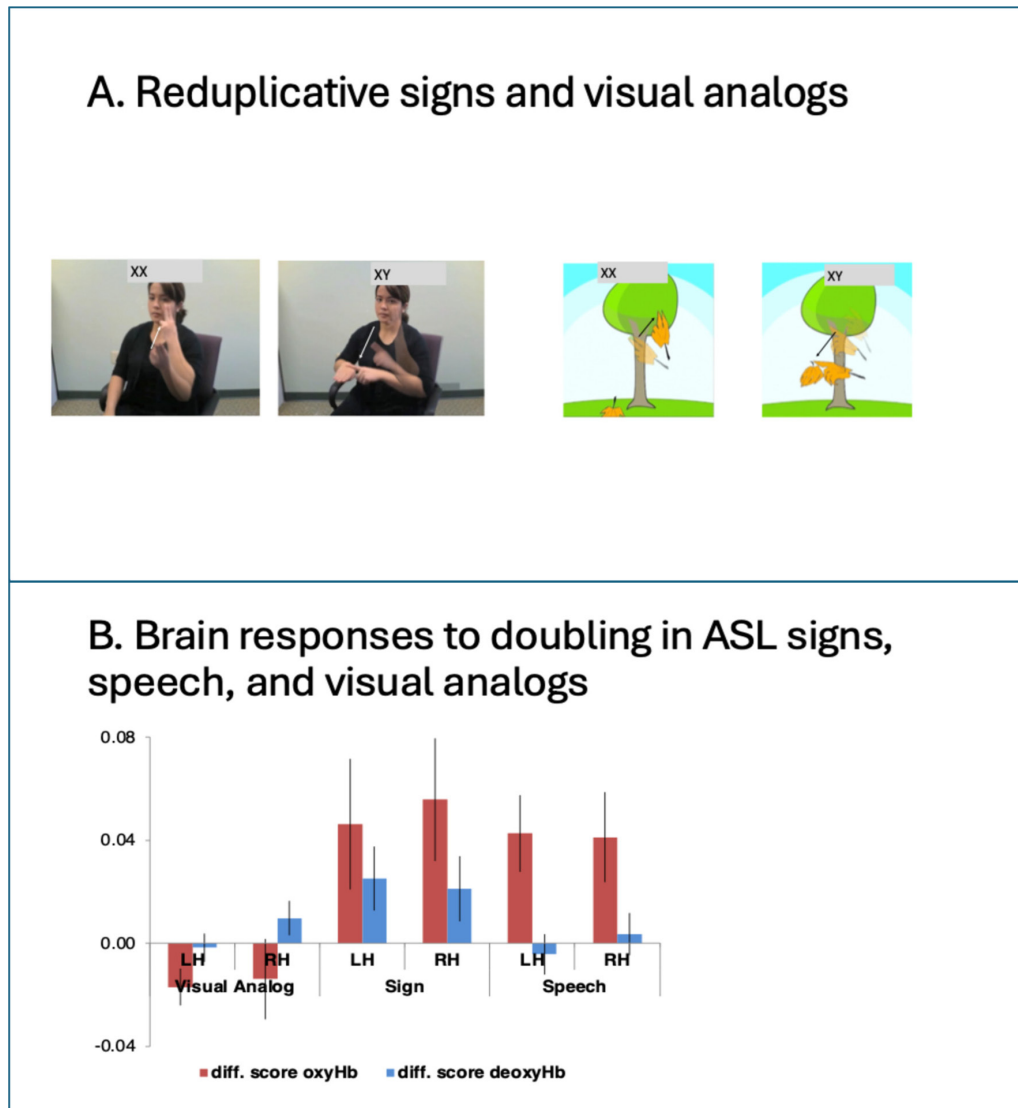


Figure 3: Amodal phonology in infants' brains. Panel A illustrates doubling in novel ASL signs and in visual analogs; Panel B provides infants' responses to doubling in signs, speech, and visual analog; means are difference scores in brain response to doubling and no-doubling; OxyHb and deoxyHb are the concentrations of oxygenated and deoxygenated hemoglobin. Figure from Berent et al. 2021b, *Scientific Reports*, used under a Creative Commons Attribution 4.0 International License.

If responses to signs only arise from their visual properties, then doubling ought to elicit similar responses in signs and visual controls; in contrast, if the infant's brain extracts linguistic structure from ASL signs, then doubling in ASL signs and controls ought to differ.

To further ascertain whether doubling in ASL signs engaged an amodal language network, the experiment also contrasted brain responses to reduplication in signs and in speech stimuli

(in another set of infants). Results showed that infants extracted the reduplicative rule (AA) from ASL signs, and the neural responses to reduplication in signs differed from reduplication in the visual controls, but were similar to the response to reduplication in speech stimuli (see **Figure 3B**; Berent et al. 2021b). These results suggest that infant brains possess a powerful phonological system that differentially responds to *all* linguistic stimuli, speech or signs.

4 Conclusions

In this piece, I outlined three arguments in support of the hypothesis that phonology is substance-free, algebraic, and amodal. First, I argued that speakers' demonstrably sensitivity to phonetic detail does not challenge the possibility that phonology is substance-free. Second, I explained why an algebraic, substance-free phonology is evolutionarily adaptive. Third, I reviewed three powerful experimental demonstrations suggesting that, indeed, phonology is abstract, algebraic, and amodal. I conclude that the phonological grammar is substance-free (Andersson 2025; Berent 2013; Chabot 2025; Gorman & Reiss 2025; Hale & Reiss 2008; Reid et al. 2017; Shen 2025; Tanaka et al. 2011; Volenec & Reiss 2020).

Competing interests

The author has no competing interests to declare.

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